

Course Code	EC24P
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Roll Number	23104B0003
Class	T.E – EXTC. SEM-V
Division	B
Date of Performance	10 th September 2025
Date of Submission	3 th October 2025

EXPERIMENT NO.10

Estimated time to complete this experiment: 02 hours.

Objective: To simulate Static CMOS and Pseudo Nmos. Perform Static Analysis, Temperature Analysis and Dynamic Analysis for both.

CO to be achieved: CO2.

Expected Outcome of Experiment:

- Students will learn cadence virtuoso.
- Students will be able to find rise time and fall time of asymmetric CMOS inverter.
- Students will be able to find propagation delay time of CMOS Inverter.

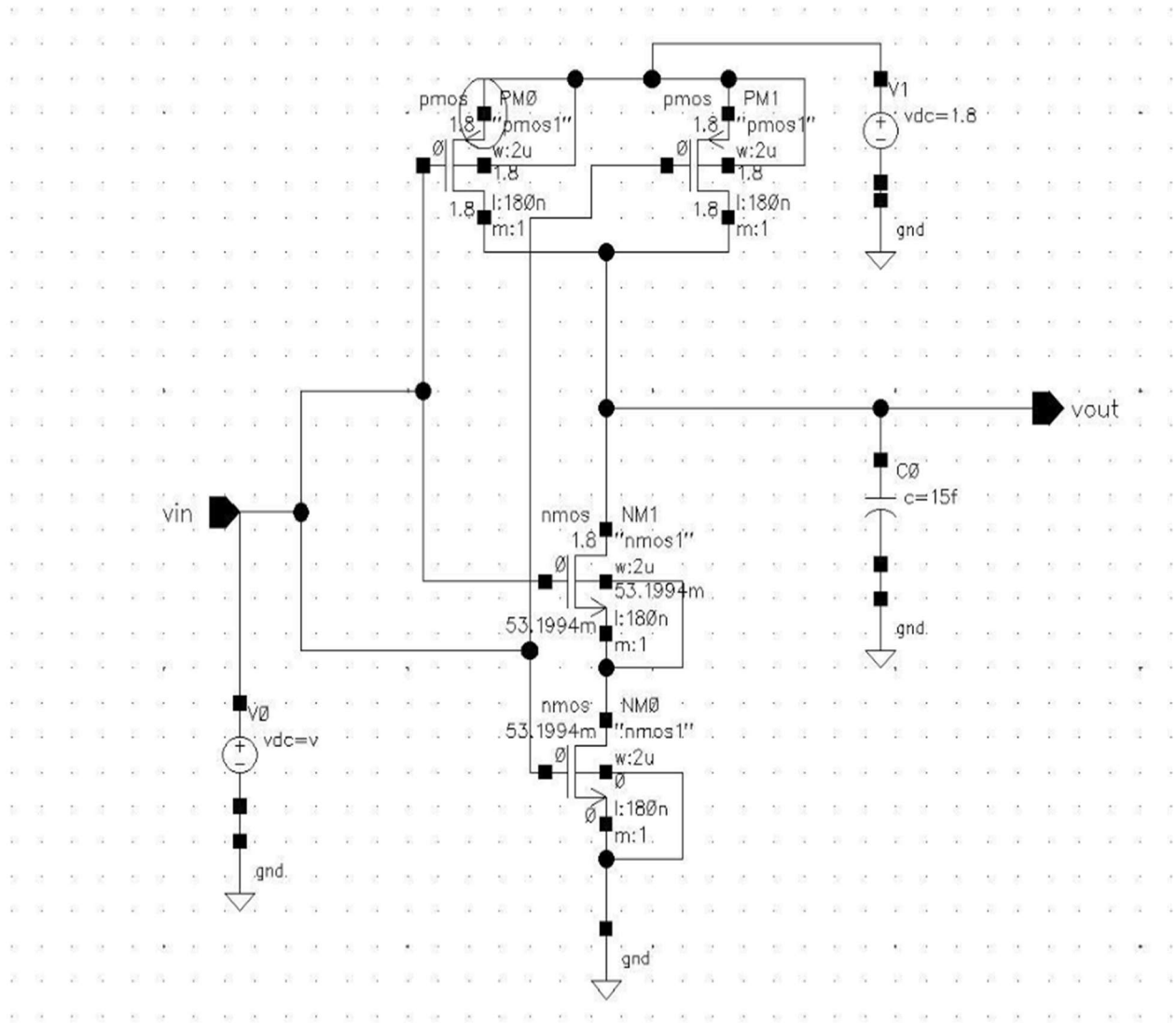
Pre Lab/ Prior Concept: Knowledge of working of MOSFET and CMOS inverter transient analysis.

Apparatus/TOOL:

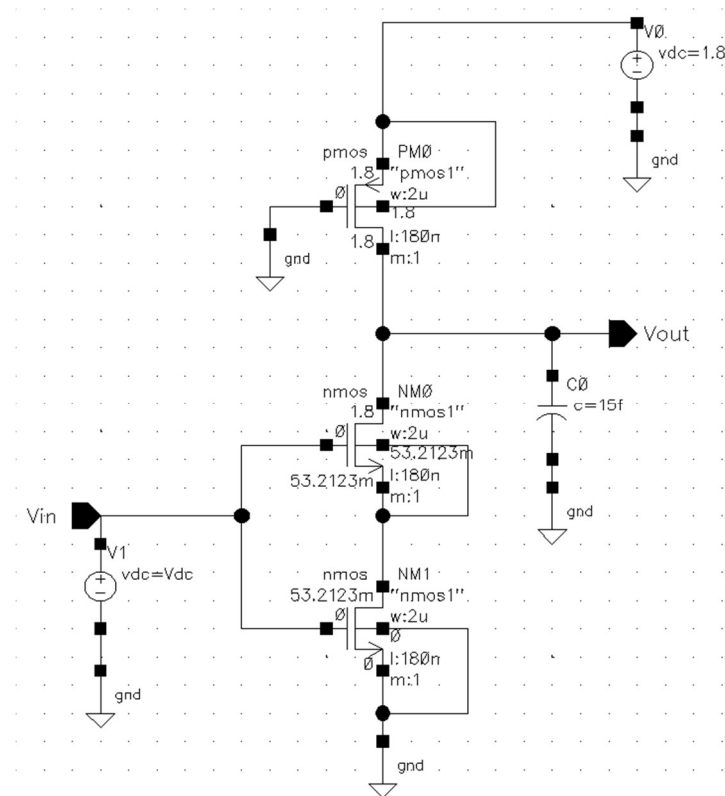
- PC
- CADENCE VIRTUOSO.

Circuit Diagram:

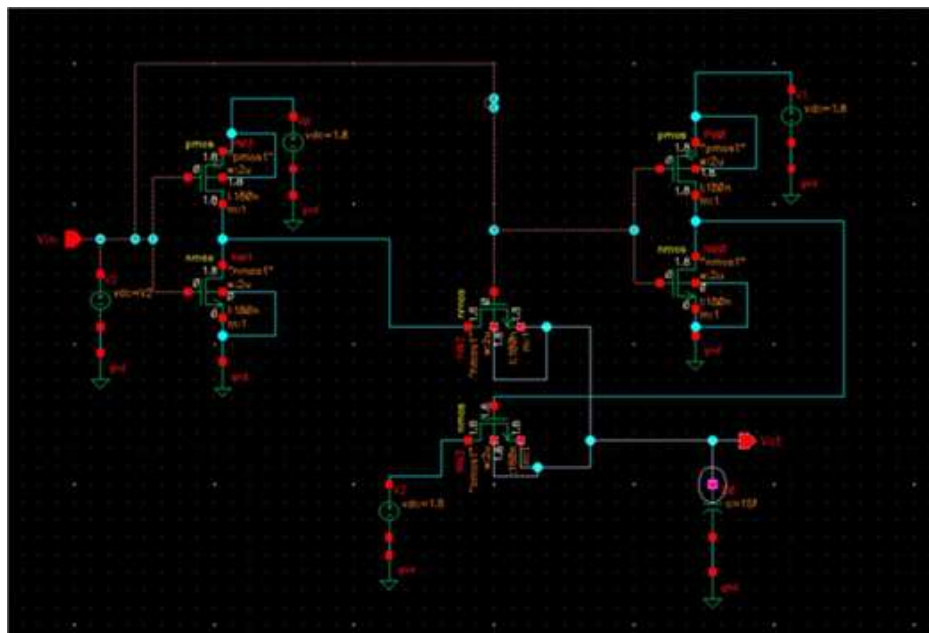
Static CMOS



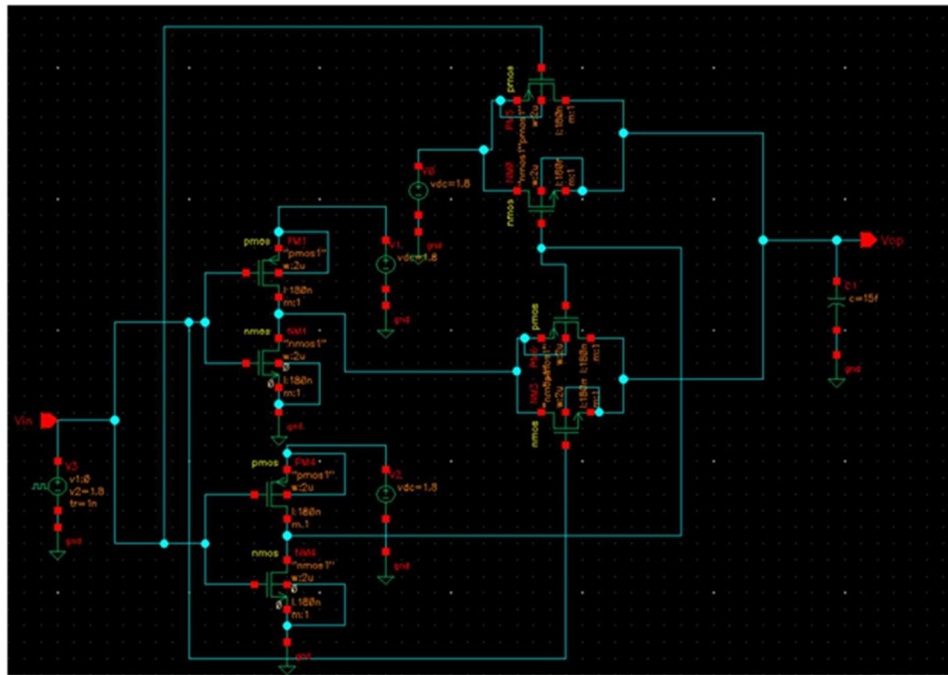
Pseudo NMOS



PTL :



TG:



Procedure

Step 1: Open Cadence Virtuoso

1. Launch Cadence Virtuoso using the following commands in the terminal:

```
csh
```

```
source /home/install/cshrc
```

```
virtuoso
```

Step 2: Create Library and Cell View

1. Create a new library using the gpdtk180 technology.
2. Create a cell view for the inverter or pseudo NMOS circuit in the new library.

Step 3: Set Circuit Parameters

1. Power Supply: Set VDD = 1.8V.
2. Input Signal (VPULSE):
 - a. Voltage 1: 0V
 - b. Voltage 2: 1.8V
 - c. Delay time: 10ns
 - d. Rise time: 1ns
 - e. Fall time: 1ns
 - f. Pulse width: 10ns
 - g. Period: 20ns

Step 4: Perform Transient Analysis

1. Set up transient analysis with a stop time of 20ns (or higher) to capture full waveform transitions.
2. Run the simulation and plot the input and output waveforms.

Step 5: Measure and Calculate Dynamic Parameters

1. TRISE: Measure the time for the output to rise from 10% to 90% of VDD.
2. TFALL: Measure the time for the output to fall from 90% to 10% of VDD.
3. TPLH: Measure the low-to-high propagation delay (time for output to go from low to high).
4. TPHL: Measure the high-to-low propagation delay (time for output to go from high to low).

Step 6: Static Analysis

1. Static CMOS:
 - a. Measure VOH (output high voltage), VOL (output low voltage), VIH (input high voltage), and VIL (input low voltage) by analyzing the voltage levels during steady-state operation.
2. Pseudo NMOS:
 - a. Perform the same voltage-level analysis for the pseudo NMOS circuit.

Step 7: Temperature Analysis

1. Set up a temperature sweep with the following temperatures: -45°C, 37.5°C, 78.75°C, 120°C.
2. Measure Vth (Threshold Voltage) at each temperature for both Static CMOS and Pseudo NMOS circuits.
 - a. Record how the threshold voltage changes with temperature.

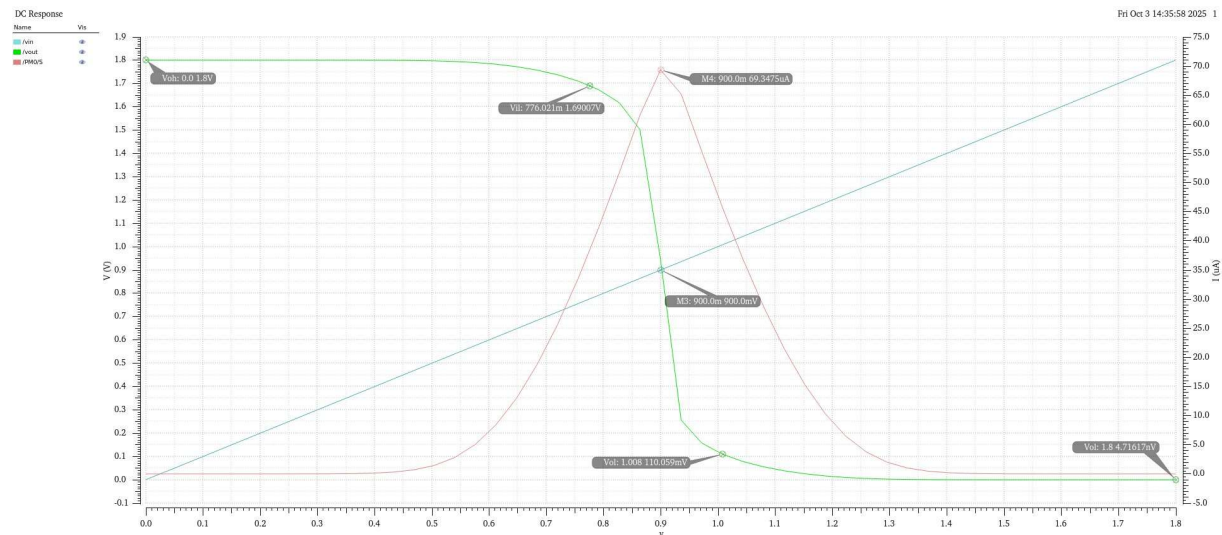
Step 8: Dynamic Analysis

1. Run transient simulation for both Static CMOS and Pseudo NMOS circuits.
2. Measure the TRISE, TFALL, TPLH, and TPHL for both circuits.
3. Compare the dynamic delays between Static CMOS and Pseudo NMOS.

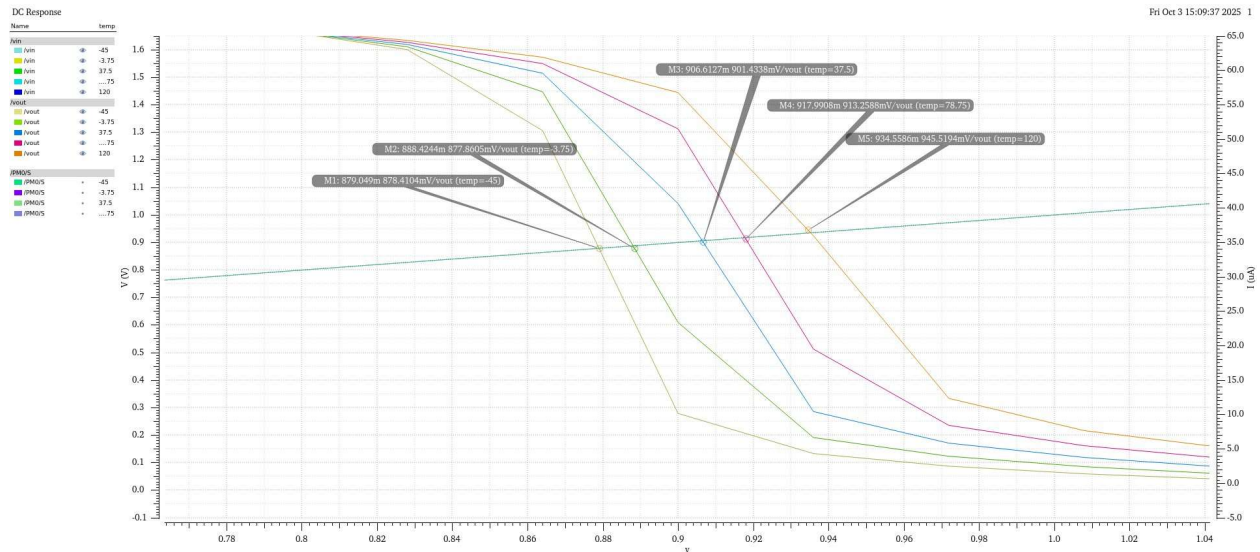
Results:

Static CMOS

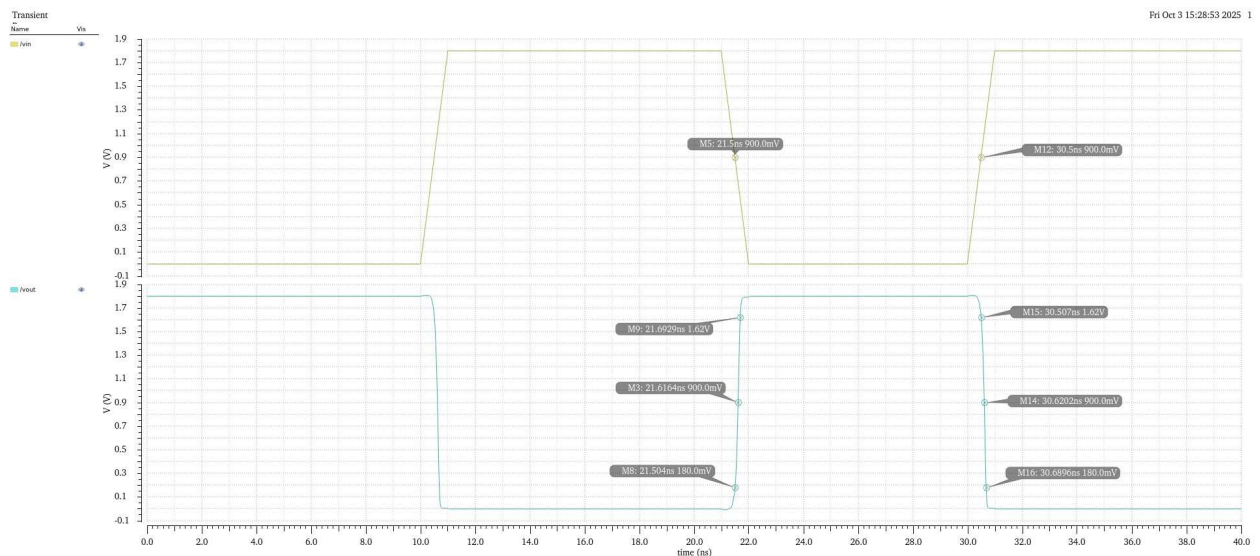
Static Analysis



Temperature Analysis

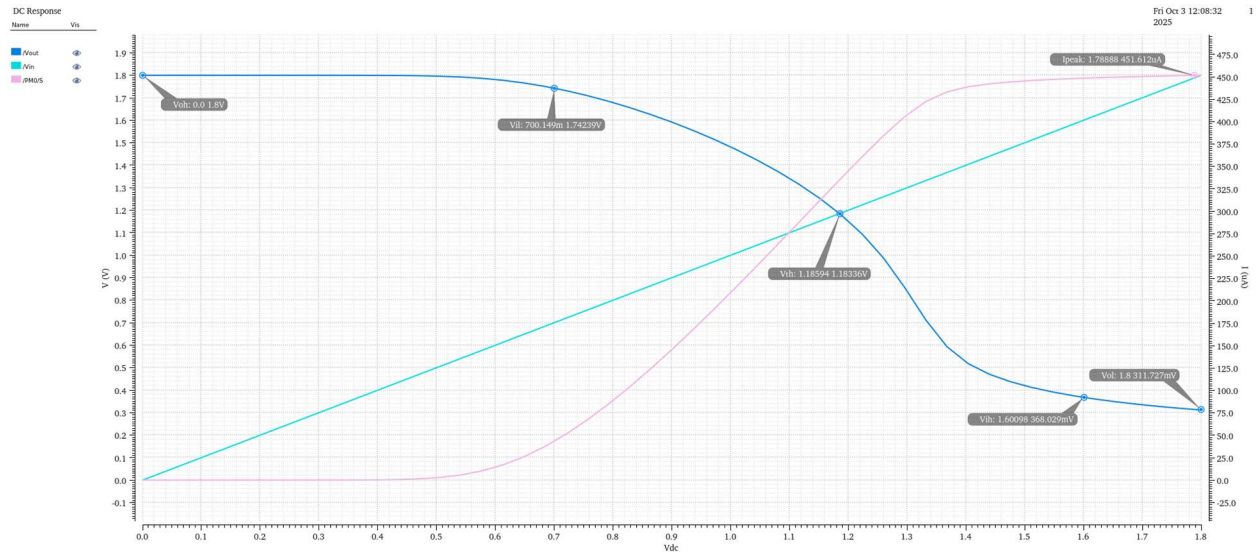


Dynamic Analysis

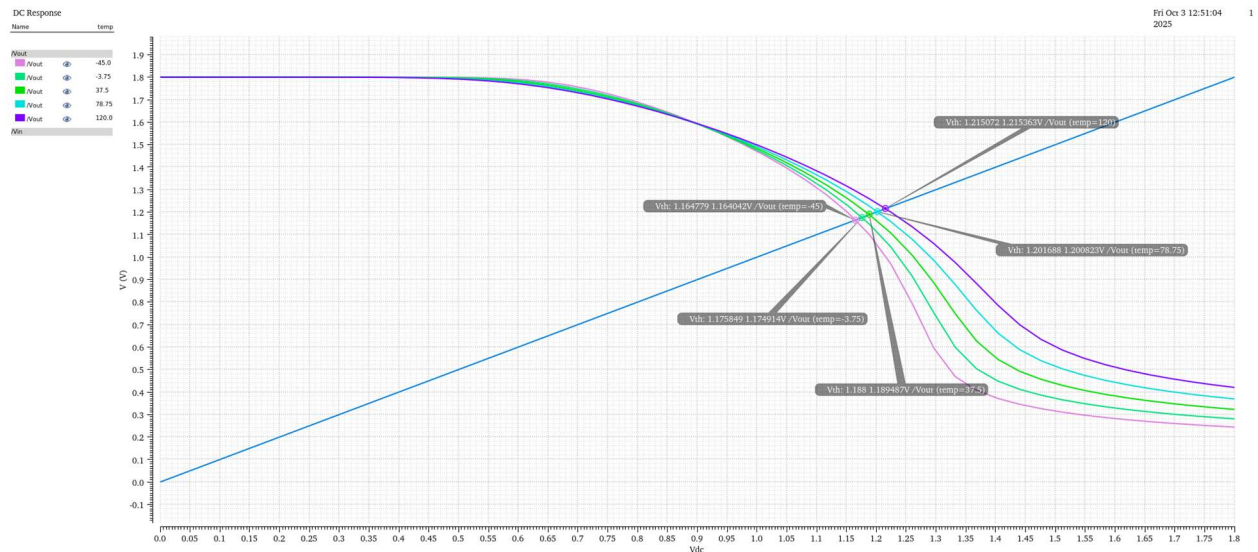


Pseudo NMOS

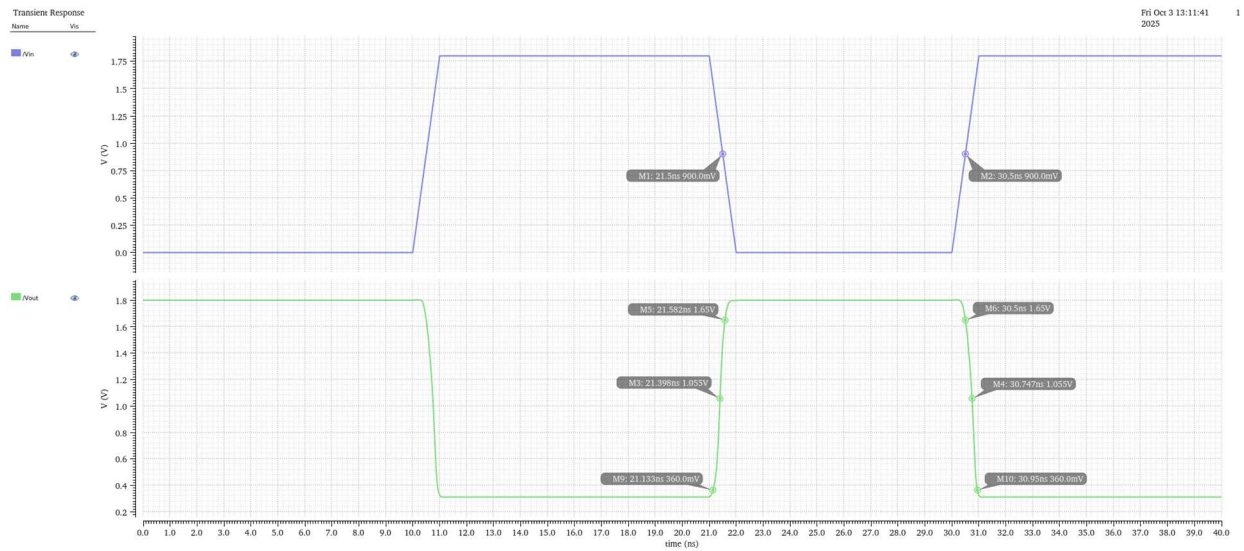
Static Analysis



Temperature Analysis

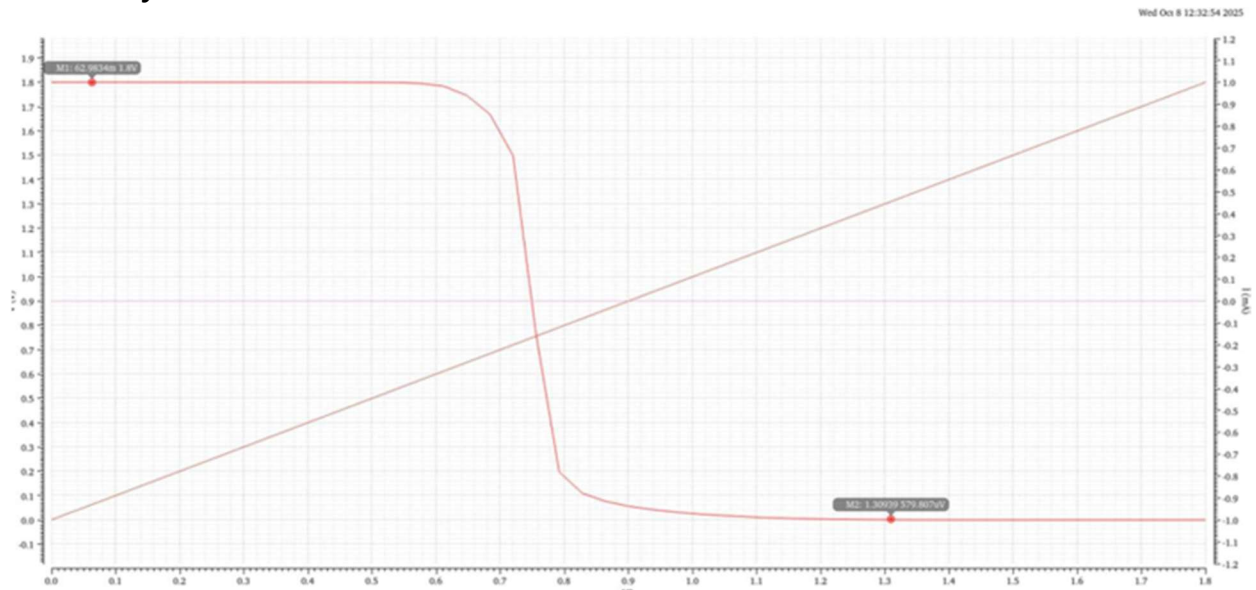


Dynamic Analysis

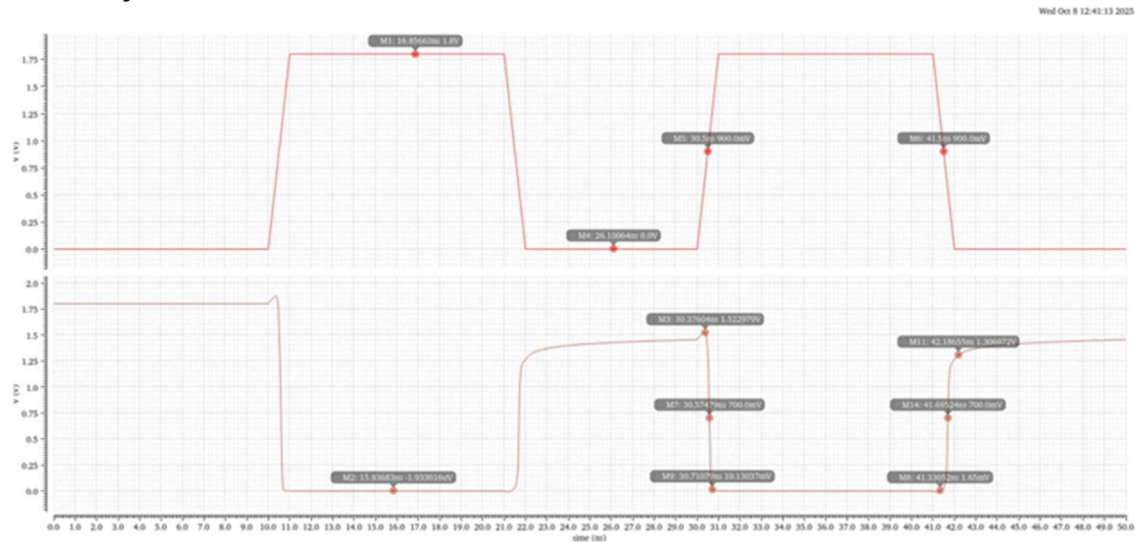


PTL:

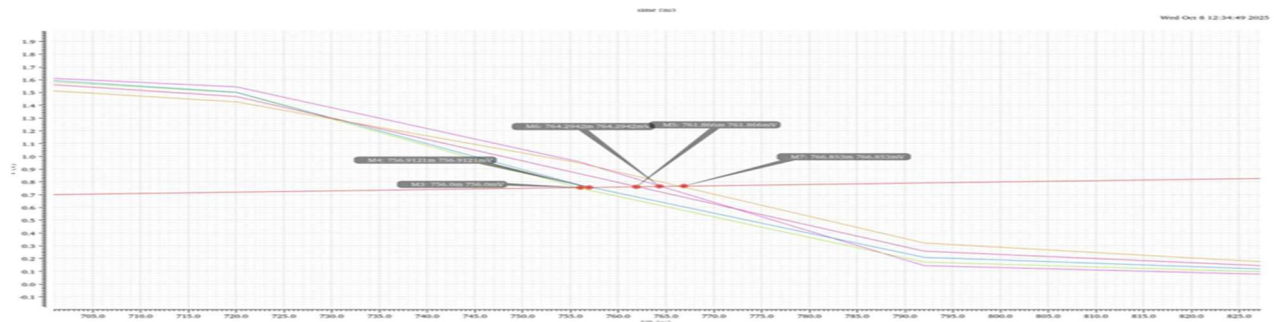
Static Analysis:



Dynamic Analysis:

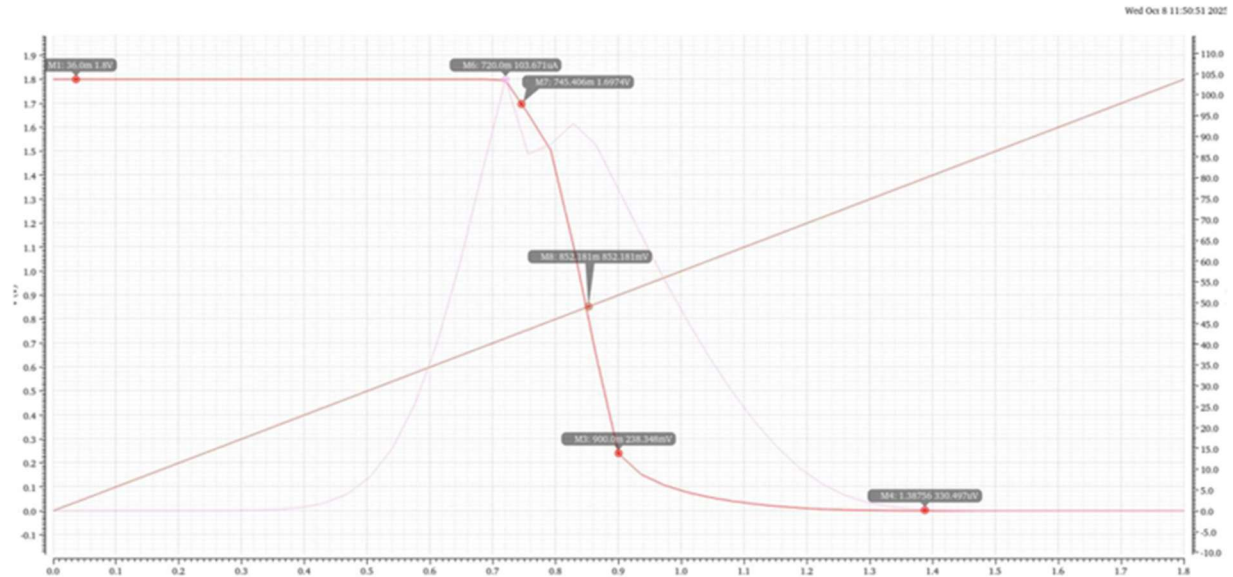


Temperature Analysis:

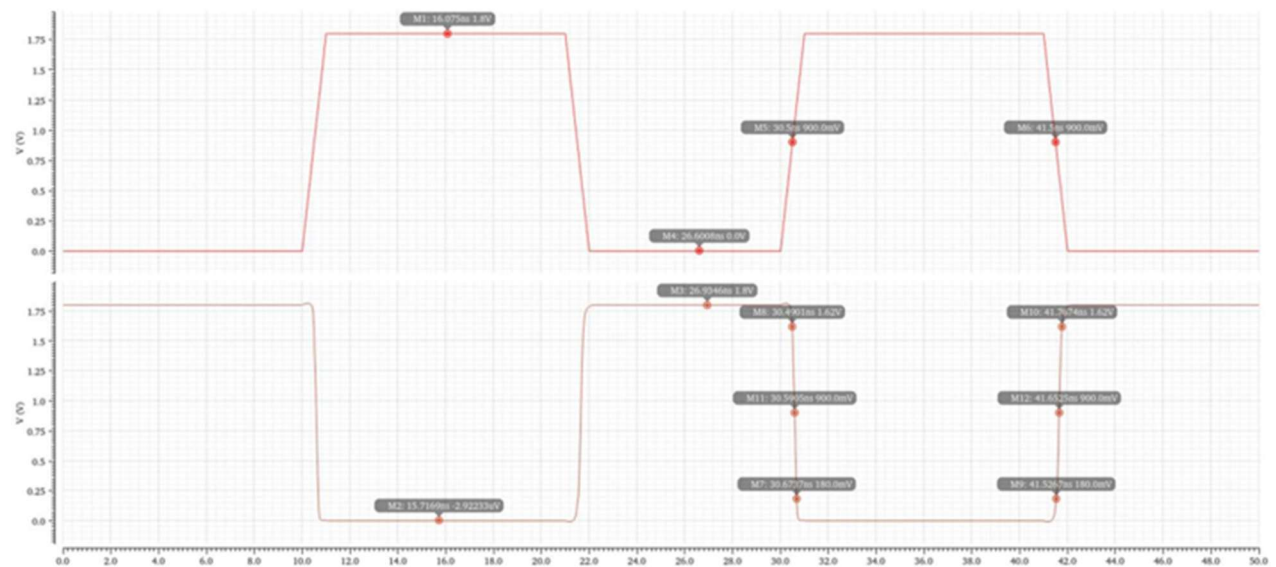


TG:

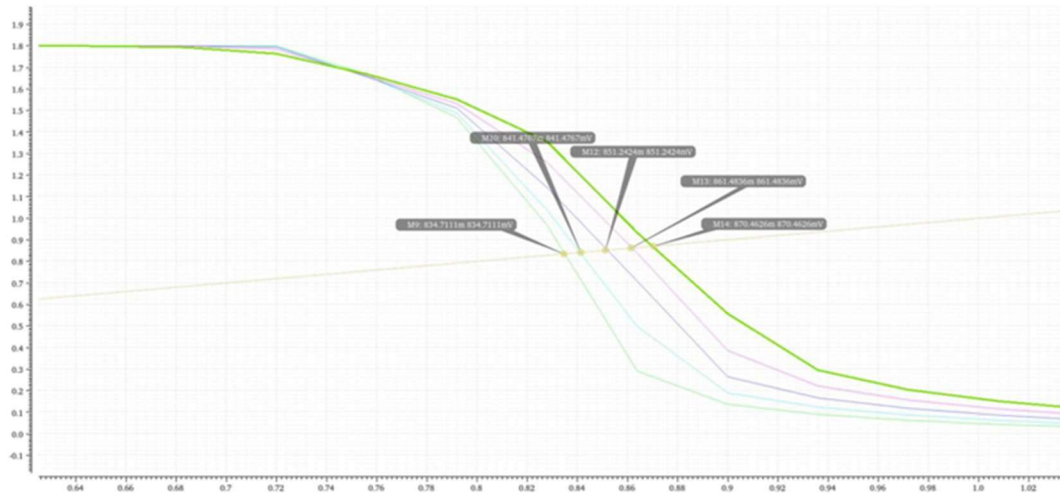
Static Analysis:



Dynamic Analysis:



Temperature Analysis:



Observation

Static Analysis	Static Cmos	Pseudo NMOS	PTL	TG
Voh	1.8V	1.8 V	1.8 V	1.8 V
Vol	4.7161 nV	311.727 mV	352n	333u
Vil	1.69V	1.7424 V	1.74	1.69
Vih	100.059 mv	368.029 mV	85.85 mV	238.34 mv
Temperature Analysis (C)	Static Cmos (Vth)	Pseudo NMOS (Vth)	PTL	TG
-45	878.4104 mV	1.164 V	763.2m	834.7m
-3.75	877.860 mV	1.175 V	755.4m	841.4m
37.5	901.433 mV	1.189 V	757.4m	851.24m
78.75	913.258 mV	1.201 V	762.3m	861.4m
120	945.519 mV	1.215V	766.3m	870.4m
Dynamic Analysis	Static Cmos	Pseudo NMOS	PTL	TG
Trise	0.1889 ns	0.449 ns	0.85	0.24
Tfall	0.18 ns	0.45 ns	0.34	0.18
Tplh	0.1164 ns	0.102 ns	0.07	0.09
Tphl	0.1202 ns	0.247 ns	0.19	0.15

Conclusion:

- **Static Analysis:**

VOH (Output High Voltage) is equal for all at around 1.8 V.
VOL (Output Low Voltage) is lowest in Static CMOS ($\sim 4.7 \mu\text{V}$), showing best low-level performance. Pseudo NMOS, PTL, and TG have higher VOL values, with TG slightly better than PTL.
VIL and VIH indicate TG and Static CMOS provide better logic-level separation and noise margins compared to PTL and Pseudo NMOS.
- **Temperature Analysis (Vth):**

Threshold voltage increases with temperature for all circuits.
Static CMOS has the lowest Vth values, showing easier switching.
Pseudo NMOS has higher Vth, requiring more voltage to turn on.
PTL and TG remain relatively stable across temperatures, with TG showing the best thermal stability.
- **Dynamic Analysis:**

Static CMOS and TG show faster rise and fall times, while PTL and Pseudo NMOS are slower.
TG has the least delay overall, followed by Static CMOS.
PTL and Pseudo NMOS exhibit higher delay values, making them slower in operation.
- **Overall Conclusion:**

Static CMOS offers good balance between speed and stability.
Pseudo NMOS reduces transistor count but has slower response and higher power.
PTL saves area and power but suffers from voltage degradation.
TG provides full voltage swing, high speed, and good stability — making it the best overall performer.

Practical Performance (5 Marks)	Presentation and Conclusion (5 Marks)	Total (10 Marks)	Sign	Date of Submission.